

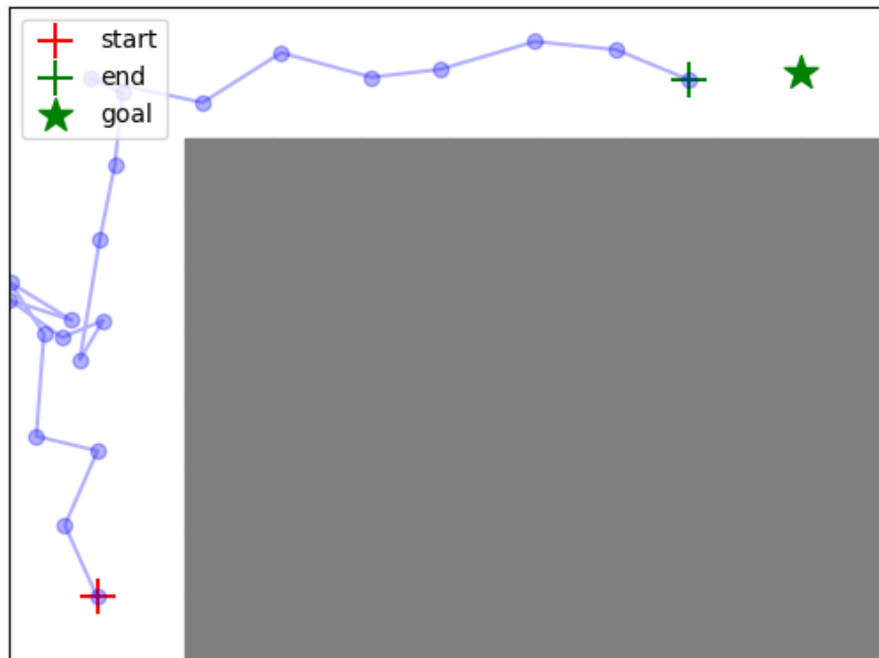
This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

1.b

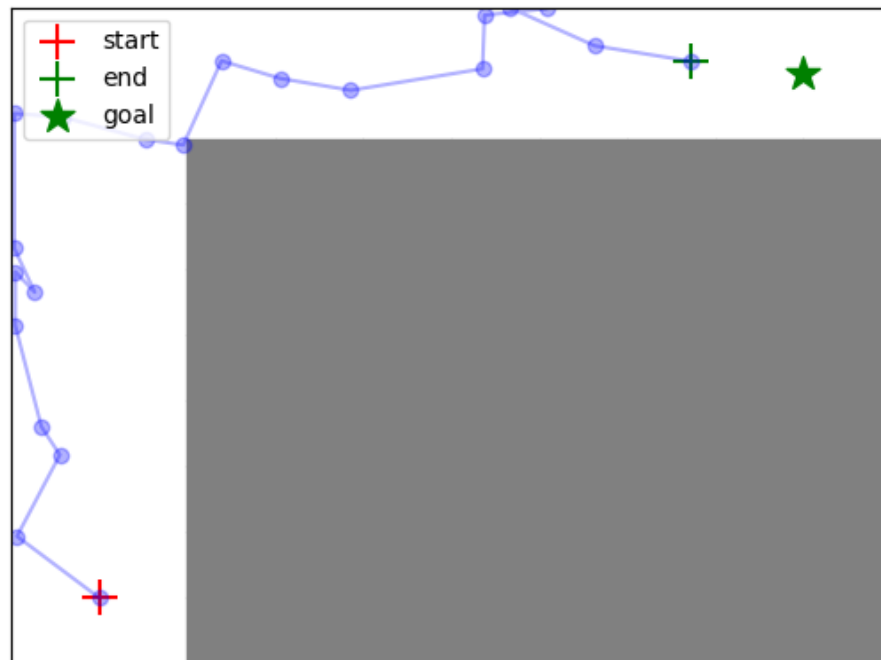
IQL Expectile	Eval_AverageReturn	Eval.StdReturn
0.2	-22.16	8.39
0.9	-19.67	5.94

Table 1: Eval Average Return for different IQL Expectile value on PointmassEasy-v0

PointmassEasy-v0 IQL Zeta=0.2 Last Trajectory Plot



PointmassEasy-v0 IQL Zeta=0.9 Last Trajectory Plot



Zeta of 0.9 outperforms zeta of 0.2. This makes sense because the stated goal of using an Expectile loss in the IQL paper is to predict an upper expectile of the TD targets that approximates the maximum of $V(s)$. With zeta of 0.2 we put a weight of 0.2 on good actions which results in a pessimistic value of $V(s)$ and therefore $V(s)$ is too low and advantages $A = Q - V$ are too high, so the actor can't distinguish good actions from bad actions because everything looks above average relative to V .

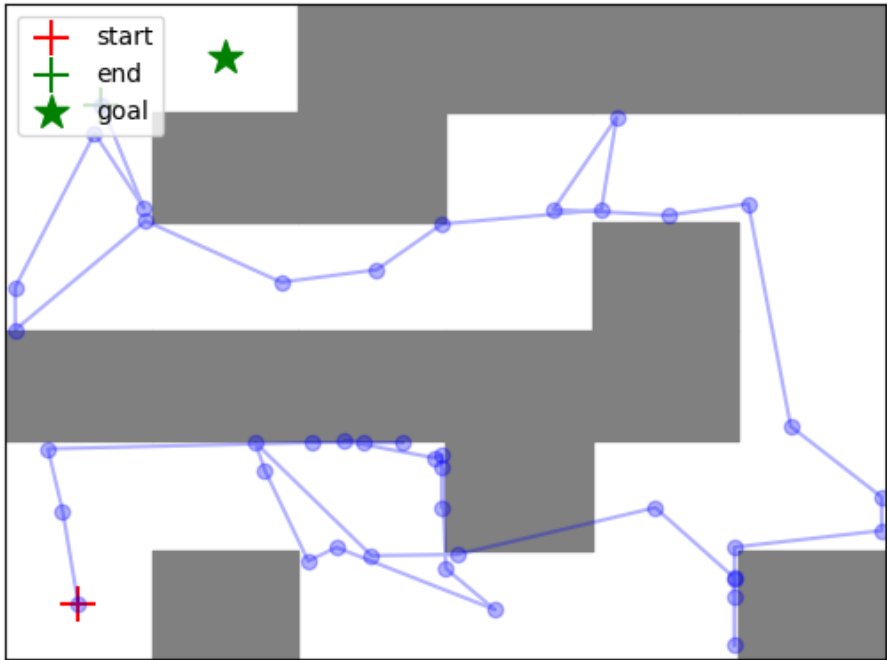
Zeta of 0.9 does the opposite, predicting an optimistic $V(s)$ for in-distribution actions, like the IQL paper intended, so the actor learns to up-weight the good in-distribution actions.

1.c

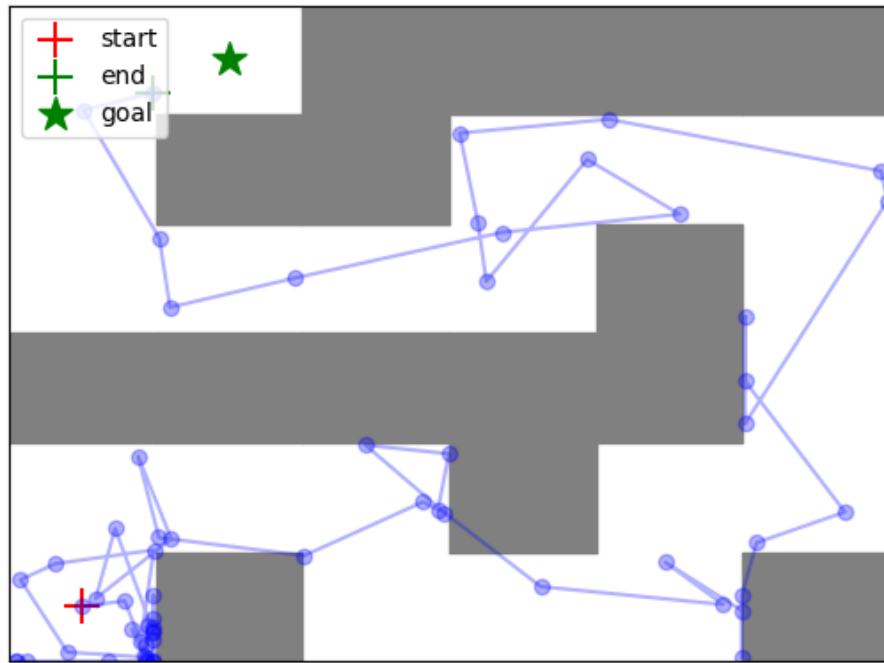
IQL Expectile	Exploration	Eval_AverageReturn	Eval_StdReturn
0.2	Random	-79.5	33.025
0.2	RND	-54.78	16.92

Table 2: Eval Average Return for different exploration strategies on PointmassMedium-v0

PointmassMedium-v0 IQL Zeta=0.2 Random Last Trajectory Plot



PointmassMedium-v0 IQL Zeta=0.2 RND Last Trajectory Plot



The offline RL policy trained on the data produced by the RND exploration strategy outperformed the policy trained on the data produced by the Random exploration strategy. This tells us that a data exploration strategy that actively seeks out novel states, rather than stumbling upon them randomly can build a better dataset for an offline RL policy.

2.b

2.c